INCH-POUND

MIL-W-19623B(SH)
11 April 1994
SUPERSEDING
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25 November 1991
(See 6.9)

MILITARY SPECIFICATION

WINDLASS: ANCHOR, ELECTRIC-HYDRAULIC, VERTICAL AND NAVY TYPE, NAVAL SHIPBOARD

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric-hydraulic, vertical shaft, anchor windlass installation on Naval ships. The windlass is used for lowering and raising an anchor and chain when anchoring the ship. The capstan heads are used for handling lines and for warping service.

1.2 Classification. The anchor windlasses are of the following classes as specified (see 6.2).

Class 1 - Vertical, electric-hydraulic, two wildcats, two capstan heads, below deck drive.
Class 2 - Vertical, electric-hydraulic, one wildcat, one capstan head, below deck drive.
Class 3 - Vertical, electric-hydraulic, two wildcats, one capstan head, below deck drive.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, SEA 05Q42, Naval Sea Systems Command, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3950

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

FF-B-171	- Bearings, Ball, Annular (General Purpose).
FF-B-185	- Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.
FF-B-187	- Bearing, Roller, Tapered.
FF-S-92	- Screw, Machine: Slotted, Cross-Recessed or Hexagon Head.
HH-L-361	- Lining, Brake; and Lining Material, Brake: Automotive.
TT-E-490	- Enamel, Silicone Alkyd Copolymer, Semigloss (for Exterior and Interior Nonresidential Use).
PPP-F-320	- Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes.

MILITARY

MIL-C-2212	- Contactors and Controllers, Electric Motor AC or DC, and Associated Switching Devices.
MIL-S-2940	- Snubbers, Fluid Pressure, Instrument Protection.
MIL-M-3184	- Machinery: Deck and Vehicle Mounted with Associated Equipment and Provisioned (Repair Parts) Items; Packaging of.
MIL-F-3541	- Fittings, Lubrication, General Specification for.
MIL-G-3859	- Grease Guns, Hand-Operated, Lever, Push and Screw Type General Specifications for.
MIL-P-15024	- Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5	- Plates, Indentification.
MIL-C-15730	- Coolers, Fluid, Naval Shipboard: Lubricating Oil, Hydraulic Oil, and Fresh Water.
MIL-B-16392	- Brakes, Magnet, Naval Shipboard.
MIL-M-17060	- Motors, 60-Hertz, Alternating Current, Integral-Horsepower, Shipboard Use.
MIL-M-17413	- Motors, Direct Current, Integral H.P., Naval Shipboard.
MIL-P-17869	- Pumps and Motors, Power, Oil Hydraulic (Naval Shipboard Use).
MIL-G-18997	- Gauge, Pressure, Dial Indicating.
MIL-L-19140	- Lumber and Plywood, Fire-Retardant Treated.
MIL-F-24402	- Filters (Hydraulic), Filter Elements (High Efficiency) and Filter Differential Pressure Indicators, General Specification for

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MIL-P-24441 - Paint, Epoxy-Polyamide General Specification for.
MIL-P-24441/1 - Paint, Epoxy-Polyamide, Green Primer, Formula 150, Type I.
MIL-P-24441/3 - Paint, Epoxy-Polyamide, Topcoat, White, Formula 152, Type I.

STANDARDS

FEDERAL

FED-STD-595 - Colors Used in Government Procurement.

MILITARY

MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-278 - Welding and Casting Standard.
MIL-STD-777 - Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

805-860303 - Hull Type Plan, Capstan and Gypsy Heads, Tabulation of Sizes.
810-1385850 - Piping Instrument, Pressure, for All Services.
53711-5381250 - Transfer Pump and Hose Assembly for Navy Standard Transmission.

(Application for copies should be addressed to: Commanding Officer, Naval Ordnance Station, Code 8022, NEDALS, Louisville, KY 40214-5001)

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

S9086-VD-STM-000/CH-631 - Preservation of Ships in Service (Surface Preparation and Painting).

(Application for copies should be addressed to the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN GEAR MANUFACTURER'S ASSOCIATION (AGMA)
STANDARDS

(Application for copies should be addressed to the American Gear Manufacturer's Association, 1500 King Street, Suite 201, Alexandria, VA 22314.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

A2.4 - Standard Symbols for Welding, Brazing and Nondestructive Examination. (DoD adopted)

(Application for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, 13th Floor, New York, NY 10036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1640 - Standard Test Methods for Drying, Curing, or Film Formation of Organic Coatings at Room Temperature.

F 765 - Standard Specification for Wildcats, Ship Anchor Chain. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

STEEL STRUCTURES PAINTING COUNCIL (SSPC)

SP 5 - White Metal Blast Cleaning.

(Application for copies should be addressed to the Steel Structures Painting Council, 4400 Fifth Avenue, Pittsburgh, PA 15213.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Material shall be as specified herein. Material not definitely specified shall be of a quality best suited for the purpose intended.

3.1.1 Cast iron. Cast iron, malleable iron, and semi-steel shall not be used for any part unless specifically approved by the contracting activity concerned for the particular application.

3.1.2 Welding. Welded fabrication may be used in lieu of castings. A full description of welding procedures shall be given using welding symbols in accordance with ANSI A2.4. Fabrication, welding, and inspection shall be in accordance with MIL-STD-278.

3.1.3 Electrical equipment. Electrical equipment shall be service "A" of the applicable electrical specifications referenced herein.

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3.1.4 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2 General design. The general design of the windlasses shall be as specified in 3.2.1 through 3.2.3 (see 6.3).

3.2.1 Class 1 windlasses. Class 1 windlasses shall consist of two wildcats, two separate capstan heads, wildcat hand brakes, electric brakes, wildcat locking heads, two electric-hydraulic power plants, change-over valve, controllers, expansion and storage tanks, bedplates, gearing, hand brake and hydraulic control stands, and all other operating and control accessories necessary for complete and independent operation of the windlasses. Wildcats and capstan heads shall be mounted on individual vertical shafts. The capstan drives may rotate with the wildcats. The electric-hydraulic power plants shall consist of two independent units located below deck in the windlass room. The units shall be operable either independently or simultaneously from the deck or in the windlass room and shall be hydraulically cross connected so that either power plant may operate either wildcat and capstan head. Wildcats and capstans shall be reversible and equally operable in either direction of rotation. Hand brake and hydraulic controls shall be pedestal mounted on the deck with a second set of controls located below deck in the windlass room.

3.2.2 Class 2 windlasses. Class 2 windlasses shall consist of one wildcat, one capstan head mounted above the wildcat, locking head, wildcat hand brake, electric brake, an electric-hydraulic power plant, expansion and storage tank, bedplates, gearing, hand brake and hydraulic control stands, and all other operating and control accessories necessary for complete and independent operation of the windlass. The capstan head shall be keyed to the vertical shaft and the wildcat shall be arranged to float on the shaft below the capstan. The electric-hydraulic power plant shall consist of one electric motor and hydraulic pump with mounting and operating accessories located below deck in the windlass room. The windlass shall be equally operable in either direction of rotation. Hand brake and hydraulic controls shall be pedestal mounted on the weather deck with a second set of controls located below deck in the windlass room.

3.2.3 Class 3 windlasses. Class 3 windlasses shall consist of two wildcats and one separate vertical capstan head driven by a single electric-hydraulic power plant located below deck in the windlass room, together with expansion and storage tank, bedplates, hand brake and hydraulic controls, locking heads, wildcat hand brakes, electric brake, gearing, and all other operating and control accessories necessary for complete and independent operation of the windlass. The two wildcats and capstan head shall be mounted on a bedplate with the capstan head centrally located above the wildcats to provide clear leads for a rope. Each wildcat and the capstan head shall be mounted on vertical shafts, the capstan drive being taken off the wildcat driving mechanism. The capstan drive may rotate with the wildcats. A separate locking head shall be provided for each wildcat. A mechanical interlock shall be provided to prevent engaging of both wildcat locking

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heads simulatenously. Wildcats and capstan shall be reversible and equally operable in either direction of rotation. Hand brake and hydraulic control wheels shall be pedestal mounted on the deck with a second set of controls located below deck in the windlass room.

3.2.4 Design detail.

3.2.4.1 Design of the windlass shall eliminate the hazard of parts becoming inoperative as a result of frequent soaking by salt spray and extreme atmospheric conditions occurring on the weather deck of the ship.

3.2.4.2 The design shall provide compensation for deck deflections not more than 1 inch in either direction. Where shafts pass through a deck, provision shall be made to obtain thorough watertightness. The equipment shall be operable and function in accordance with all requirements of this specification when the ship is rolling or pitching in a sea or permanently inclined up to 15 degrees from the vertical.

3.2.4.3 The windlass shall be as compact and light in weight as is consistent with adequate strength, rigidity, and durability to withstand stresses encountered in windlass operation. If required, maximum size and weight shall not exceed that specified (see 6.2).

3.2.4.4 Positive means shall be provided for securing keys, feathers, splines, nuts, screws, bolts, and collars to prevent their working loose. The use of set screws shall be avoided. When set screws are used, means shall be provided to prevent their loosening and working out.

3.2.4.5 Each power plant shall consist of an electric motor driving a variable stroke hydraulic pump. A hydraulic motor driven by the hydraulic pump, shall drive, through reduction gearing, the vertical wildcat shaft and the capstan shaft.

3.2.4.6 All handwheels, levers, and associated members shall be designed to withstand a force of 350 pounds, applied to the rim or handle when the mechanism is driven against its stops or extreme position, without failure or permanent deformation.

3.2.4.7 When specified (see 6.2), a speed limiting device shall be furnished. This device shall be positively coupled to the wildcats and shall be designed to limit the speed at which the anchor could be lowered under any conditions to a maximum of 1,000 feet per minute. During the free fall condition, the speed control device shall maintain the speed of the anchor descent to within a range from 800 feet per minute to 1000 feet per minute. A device shall be provided to monitor the chain speed at all times.

3.2.5 Wildcats and capstan heads. The wildcats shall be designed to handle flash butt welded stud link chain and its connecting links of the size specified (see 6.2). The wildcat shall be designed in accordance with the ASTM F 765, type IV. The contractor shall furnish and use 5-link and 3-link sections (total eight links) of the specified size of chain and one detachable link for testing and assuring the fit of the chain on the wildcat casting. The 5-link and 3-link sections shall be removed from the bitter end of the anchor chain. The 5-link and

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3-link sections shall be reinstalled on bitter end of anchor chain after completion of wildcat fit check by use of detachable links furnished by the contractor.

3.2.5.1 Each wildcat shall be arranged for positive drive through a driving head and locking head device located in the windlass room and having two or more sliding block keys for engaging the heads one with the other. The sliding block keys shall be actuated together by a common operating mechanism of rugged construction which shall positively hold the keys in the engaged or disengaged position. The entire operation of engaging or disengaging the wildcat drive shall be accomplished by one motion of a lever through an angle not exceeding 60 degrees. The operating gear and lever shall be conveniently located in the windlass machinery room.

3.2.5.2 The capstan head may turn with the wildcat, however, the capstan shall be operable with the wildcat stationary.

3.2.5.3 Capstan heads shall be of rugged construction and proportioned in accordance with Drawing 805-860303 for the rope size specified (see 6.2).

3.2.5.4 The tops of the wildcats and capstan heads shall be designed with a step recess on the periphery and shall be fitted with a convex cover plate having a rim that fits in the step recess. Cover plate shall be of corrosion-resistant metal, bronze or stainless steel, and shall be secured with round head machine screws in accordance with FF-S-92 and gasketed to prevent moisture entering into the interior of the hardware.

3.2.5.5 Both the wildcat and capstan shall be mounted on suitable bearings so supported that the drive shafts are not subjected to bending, the chain and rope pull operating stress and forces shall be completely and adequately taken by the deck bedplates.

3.2.5.6 Wildcats and capstan heads shall be shrunk on their respective barrels or shafts and keyed with keys having a drive fit. Keyway corners shall be designed with fillet radii to prevent local stress concentration by operating forces.

3.2.5.7 Chain strippers shall be provided for freeing the chain from the wildcats when heaving in or veering out.

3.2.6 Shock requirements.

3.2.6.1 The graphs (see figure 1) of shock design factor, together with qualifying notes, shall be used for design of bases, holding down bolts, and fittings. This information does not apply to components for which the high-impact (H.I.) shockproof requirements are specifically set forth by the applicable specification.

3.2.6.2 The bases, holding down bolts, and footings of all components shall be designed to resist shock by using block design factors (see figure 1) for determining shock forces. Determinations of stresses resulting from these forces are accomplished by application of current methods of analytical mechanics. Allowable stress in components due to shock shall be the yield-strength (0.2 percent offset) of the material. The stresses calculated on the basis of shock

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design loads shall be combined with the stresses encountered under normal operating conditions. Tensile stress in a bolt due to initial pull-up need not be considered. It is of paramount importance that the shock design factor shown on the curves be utilized in the design of hold down bolts, supporting feet or lugs, and main structural members of equipment, and assures that no permanent deformation or failure occurs as a result of shock. For assemblies, the shock design factors used in design of the base, holding down bolts, and footings for the entire assembly shall also be used in designing the hold down bolts and footings of the individual components unless the designer's knowledge indicates that a higher shock design factor may be required for the components, and can be supported by calculations and test data.

3.2.6.3 Bolts designed to be stressed in shear shall be installed in holes with a minimum of clearance.

3.2.6.4 Items of equipment and machinery shall not be rigidly attached to two structures which can deflect relative to each other under shock loadings. For the purpose of this paragraph, structures are defined as bulkheads, decks, and the hull of the ship. Where sway braces must be employed to afford stability, the sway brace shall be designed to fail or permit large deformations under a load caused by a force of five times the weight of the equipment and applied at the center of gravity of the item.

3.3 Capacities. The capacities of the windlass equipment shall be as follows:

3.3.1 Without exceeding the full load rating of the motor, the windlass shall hoist an anchor and chain of the specified weight and size (see 6.2) from the depth of 60 fathoms at an average rate of 6 fathoms per minute and from depth of 100 fathoms up to 60 fathoms at no specified speed. The design of class 1 windlasses shall be such that both anchors may be hoisted simultaneously.

3.3.2 Capstan heads on class 1 and class 3 windlasses shall develop a rope speed of at least 125 feet per minute with the specified size of rope at no load condition. Capstan heads on class 2 windlasses shall develop a minimum speed of 50 feet per minute. The capstan and its driving and mounting equipment shall be capable of withstanding heaving loads up to the relief valve setting of the hydraulic power unit. The capstan shall also withstand a static load equal to the minimum breaking strength of the specified size of rope using the head as a bitt at the mid-height of the head and normal to its axis. Under either load, the combined stresses in any part of the equipment shall not exceed specified allowable stresses (see 3.5.2 and 3.5.3).

3.3.3 The working pressures in the hydraulic system shall not exceed those recommended by the manufacturers of the hydraulic units.

3.3.3.1 Hydraulic units shall conform to MIL-P-17869, shall not be operated above their designed speed, and shall have sufficient capacity to meet all specified requirements.

3.3.4 Brakes shall be of ample size and strength to permit performance of specified brake tests without combined stresses in any part of the brake or operating mechanism exceeding the allowable stresses (see 3.5.4).

3.3.5 Expansion and replenishing tanks for the hydraulic fluid shall have

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a capacity of at least 125 percent of the capacity of the hydraulic system, including fluid, to the working level in the tank.

3.3.6 When hoisting an anchor and chain from a depth of 60 fathoms at the designed normal working pressure, the combined stresses in the equipment shall not exceed 35 percent of the tensile yield point of the materials used (see 3.5.1).

3.4 Detail requirements.

3.4.1 Hand brakes. Each wildcat shall be fitted with an externally contracting, hand operated, band brake located below the weather deck and controlled by a screw-type handwheel. The brake shall stop and hold the anchor and chain when dropped free from a depth of 45 fathoms to 60 fathoms. The brake shall stop and hold the anchor and chain in not more than 4 fathoms of travel. Windlasses for handling anchors weighing 30,000 pounds and over or designed for anchor chain of 3-1/2 inch wire diameter and larger shall be provided with hydraulic power assist control in addition to manual screw-type handwheel control for applying the wildcat brakes. The hydraulic power assist control shall be located on the manual handwheel control. Each wildcat brake control hydraulic power assist or manual screw-type handwheel, when selected by the operator, shall apply the wildcat brakes in all modes of windlass operation without requiring any change or adjustment to windlass components.

3.4.1.1 The hand brake bands shall be lined with non-asbestos, nonburn, brake band lining in accordance with HH-L-361. Unit pressure for brake lining shall not exceed that recommended by the brake lining contractor.

3.4.1.2 Brakes for all class windlasses shall be designed to meet all requirements with the wildcat turning in either direction.

3.4.1.3 The hand brakes shall be operated by horizontal handwheels, one for each wildcat located above the weather deck and one in the windlass room below deck. The brake mechanism shall be quick-acting and shall be such that the handwheels will supply the necessary torque without the use of levers or other equipment.

3.4.1.4 The effort required at the rim of the brake handwheel to set the brake shall not exceed 100 pounds under any of the specified test conditions.

3.4.1.5 Means shall be provided for lifting and keeping brake bands free of their drums when brakes are released. Necessary supports and guides to ensure smooth and easy operation of brake bands and operating mechanism shall be furnished.

3.4.1.6 The deck hand brake control handwheels shall be horizontally mounted on pedestals at a convenient height above the deck for operating personnel and located with full regard to protect operating personnel from the whip of the bitter end of the chain.

3.4.1.7 Brake handwheels shall be of adequate diameter, shall turn clockwise to set the brake, and shall be suitably marked to indicate the directions of rotation for setting and releasing the brake. The number of turns from set position to complete off position shall not exceed three turns. The brake

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operating gear and mechanism shall be nonoverhauling and self-locking. Indicator showing "off" or "on" position shall be provided at each control station.

3.4.1.8 Brake control rod joints shall be such that mating components shall be secured with two keys 90 degrees apart. The keys shall be secured positively with self-locking set screws or by other means.

3.4.2 Gearing. Gears shall be in accordance with American Gear Manufacturers Association standards and be designed to transmit the full torque of the windlass. Gear capacity in relation to beam strength and wear shall be computed in accordance with AGMA standards.

3.4.2.1 Gears shall have machine-cut teeth, carefully aligned and mounted.

3.4.2.2 Gears shall be of all steel construction, either forged or cast. Large gears may have welded rims on cast or forged hubs.

3.4.2.3 Gear shafts shall be designed for minimum deflection. Gears, except small spur gears, shall be held axially on their shafts by positive means in addition to a press fit. Snap rings are not acceptable as thrust collars.

3.4.2.4 Gearing shall be fully enclosed in an oil-tight case. Gearing shall be pressure lubricated from a power-driven oil pump or by suitable splash system for lubrication of gears with pressure lubrication of upper bearings. A strainer shall be provided in a power-driven oil pump lubrication system.

3.4.2.5 Gear cases shall be arranged for ready removal of gears and bearings. Cases shall be provided with bolt or screw fastened oiltight inspection covers and fill and drain openings equipped with oil sample valves. Oiltight inspection covers shall be provided in sufficient number to allow visual examination of the teeth on all gears, permit cleaning of the gear case bottom and inspect oil lubricating pump strainers. Cases shall have means for determining and limiting the oil level within the case without removing the cover. Recommended oil level shall be indicated on a level indicator dipstick or sight glass. An air vent, suitably baffled to prevent loss of oil under operating conditions and of a type which will exclude foreign matter, shall be provided for each gear case. Lifting eyebolts shall be installed as necessary on gear cases. Gear case joints shall be oiltight.

3.4.3 Bedplates. Bedplates shall be of cast or welded steel construction of adequate rigidity and strength to maintain alignment of all parts and to transmit external forces to the ship's structure without objectionable deflection and also to resist shock forces as specified in 3.2.6. Bedplates to be installed on weather decks or in the anchor windlass room shall be watertight with no pockets or hollows in which water or debris may collect. The underside of bedplates shall be finished to a plane surface for watertight mounting on foundations.

3.4.3.1 Each wildcat shall be mounted on a bedplate which may be in one piece or two pieces adequately bolted together. Where required (see 6.2), this bedplate shall contain the chain fairleads to the wildcat from the hawse pipe and the chain pipe. Fairleads shall be formed to an easy radii and shall take not less than three complete links of the chain. Heights of fairleads and chain pipe deck bolsters shall be such as to deliver the chain to the horizontal center line

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of the wildcat between the whelps. Location of chain pipe deck bolsters shall be suitable for handling chain in or out. The angle of rake and details for connection of the chain pipes to chain pipe deck bolsters will be furnished to the contractor upon request to the contracting activity after award of contract. Radii of flared mouth of chain pipe deck bolsters leading to wildcats shall be as large as practicable. Chain pipe deck bolsters where required (see 6.2), shall be fitted with suitable portable watertight metal bucklers fastened by means of hinged drop bolts with nondetachable bronze nuts.

3.4.3.2 Each power unit consisting of electric motor, reduction gearing, if required, hydraulic pump and control equipment together with all accessories shall be mounted on a bedplate in the windlass room. Bearing pads for units mounted on bedplates shall be machined to flat surfaces and fitted bolts shall be used for securing the units to bedplates. Tap bolts shall not be used unless specifically approved by the contracting activity concerned.

3.4.4 Power transmissions. High speed gear reduction between the electric motor and the hydraulic pump, if required, shall be of balanced helical or herringbone type, mounted on antifriction bearings, fully enclosed, and positively lubricated.

3.4.4.1 Hydraulic units shall be of the multiple piston type, of standard size in accordance with MIL-P-17869. Only units which are listed on the applicable Qualified Products List on the date of invitation for bids or which may be added to that Qualified Products List subsequent to that date shall be utilized in the construction of equipment specified to be in accordance with this specification. In the event no Qualified Products List has been issued, the contractor shall request instructions as to what testing will be required to determine whether the product meets the requirements of the specification.

3.4.4.2 Flexible all-steel couplings shall be fitted between the electric motor and the high speed reduction gears unless the gears are mounted within the electric motor casing. Where no high speed reduction is required, the coupling shall be between the motor and the pump. All-steel flexible couplings shall be provided between the hydraulic motor and the drive shaft of the main reduction gear. The coupling arrangements shall be such that any hydraulic unit may be removed with a minimum disturbance of other equipment.

3.4.4.3 Slip joints or couplings shall be provided for vertical drive shafts where deflection or warping of decks might cause misalignment of the shafts.

3.4.5 Bearings.

3.4.5.1 Antifriction bearings shall be used throughout where gear alignment is involved. Antifriction bearings shall be in accordance with FF-B-171, FF-B-185, or FF-B-187, as required.

3.4.5.2 Wildcat and capstan head bearings may be bronze bushed. Bearing pressures shall be sufficiently low to assure long life without replacement.

3.4.5.3 Bronze fitted bushings and thrust washers that are pierced by oil or grease holes shall be positively fastened to prevent their turning and cutting off lubrication.

3.4.5.4 Special effort shall be made to ensure that shaft deflections

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likely to occur will not damage antifriction bearings. Bearings should be located as close to load on the shaft as practicable.

3.4.5.5 The use of snap rings to restrain bearings or to take bearing thrust is not acceptable.

3.4.6 Lubrication.

3.4.6.1 Suitable means for adequate lubrication of all working parts from readily accessible locations shall be provided. If a power-driven oil pump is provided, then it shall be furnished for spray type lubrication of gears and bearings of the main gear drive between the hydraulic motor and the wildcat and capstan shafts.

3.4.6.2 Pressure grease lubrication in accordance with type I of MIL-G-3859 shall be provided for all other bearings. Also, all bearings in gear housings not adequately lubricated by oil in the housing shall be fitted with grease fittings for pressure grease lubrication in accordance with MIL-F-3541, type I, III, IV, or V. Large bearings, such as wildcat and capstan head bearings, shall be provided with several lubricant fittings but not less than three fittings equally spaced on the periphery of the bearing. Each lubricating fitting shall be installed at safe distance from rotating components of the anchor windlass and, if possible, at one common place.

3.4.7 Controls.

3.4.7.1 Controls shall include equipment for remote operation of valves, brakes, and hydraulic pump stroke. Controls shall be arranged for convenient and safe operation of the equipment. Controls shall remain in the position set by the operator.

3.4.7.2 The windlass shall be started and stopped by electric master switches on the weather deck and in the windlass room. Control of the windlass speed shall be by hydraulic control handwheels on the deck and in the windlass room.

3.4.7.3 The hydraulic control, one for each power plant, shall include all handwheels, shafting, pedestals, gearing, bearings, and supporting brackets necessary for operating the windlass from the weather deck or windlass room. The control shall include an adjustable and self-releasing centering device at each pump for placing the hydraulic pump in neutral or no-stroke position and for adjusting the control to prevent creepage of a load.

3.4.7.4 Control leads shall consist of suitable joints, shafts, and gears mounted on antifriction bearings and arranged for minimum lost motion. Bearings and gears shall be completely enclosed. Bearings shall be spaced and located to prevent whipping and excessive deflection of control shafting. Steel tubing may be used for shafting where practicable to save weight and secure torsional stiffness. Universal joints or splined slip joints shall be provided where necessary to allow for deflection and working of ship structure. Universal joints shall be secured to control shafts with keys locked by self-locking set screws or by other means. Stuffing boxes shall be furnished where control shafting passes through

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watertight decks or bulkheads. Roll pins, tapered pins, and bolts shall not be used in joint of control shafting.

3.4.7.5 Antifriction bearings shall be used for control shafting. Gears shall be accurately cut and set up with a minimum of backlash.

3.4.7.6 Hydraulic control handwheels shall turn in the direction of wildcat or capstan operation in normal and cross connect modes of operation. Handwheels shall be marked to indicate hoist-lower on the direction of rotation. Handwheel markings to indicate percentage of pump stroke shall be provided at operating stations.

3.4.7.7 Hydraulic control handwheels shall require not more than 30 pounds effort at the wheel rim for normal operation. The turns from zero stroke to full stroke should not exceed 1-1/2 turns.

3.4.8 Piping, valves, and hydraulic accessories.

3.4.8.1 The windlass contractor shall provide fittings for each windlass, including pressure gauges, valves, relief valves, cocks, union couplings, companion flanges, gaskets, flange bolts, air vents, drain valves, and other items necessary for complete hydraulic piping system as approved except for the connecting piping and supports which are not an integral part of the equipments. Connecting piping, tubing, and supports shall be furnished by the windlass installing activity.

3.4.8.2 High-pressure connections, except those for pressure gauge connections, shall be flanged companion type with male and female joints. Flanges should be provided with full copper-covered non-asbestos gaskets, the inside diameter of which shall be such that no restriction of oil passage shall occur when gaskets are compressed. The copper-covered non-asbestos gaskets shall be single copper-covered type with copper inlay or the double jacketed type. O-rings may be used in lieu of gaskets provided equal performance is assured.

3.4.8.3 Low-pressure connections shall be silver-brazed, welded unions or flanges and shut-off valves shall be provided in hydraulic lines to permit renewal of broken lines without drainage of the main hydraulic system.

3.4.8.4 Pressure gauges in accordance with MIL-G-18997 shall be provided in servo, replenishing, and main pressure supply and return lines. If hydraulic power assist is provided to wildcat brakes, this system shall contain a pressure gauge. Gauge installation shall be in accordance with Drawing 810-1385850 and shall include a snubber in accordance with MIL-S-2940 for each gauge.

3.4.8.5 Class 1 windlasses shall be provided with a multiport changeover valve, or valves, having positive axial adjustment and hydraulic balance, for installation in the pressure piping between the hydraulic motors and pumps to permit driving either wildcat by either pump and automatically, hydraulically short circuit the idle pump. The valve should be manually operable by one man but power operation shall be provided if the force necessary to shift the valve exceeds 100 pounds; means for emergency hand operation shall be furnished in event of power failure. The changeover valve shall be located so that relief and replenishing valves are installed between the changeover valve and hydraulic pumps. Cross connecting valve shall have provisions for securing by padlock in

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the normally closed position. Caution plates shall be provided to caution against cross connecting units after failure due to contamination.

3.4.8.6 An expansion and replenishing tank shall be provided for class 2 and 3 windlasses with a capacity of at least 125 percent of the hydraulic system capacity and located above the highest point of the main pressure piping. Tanks shall be of riveted and sweated brass or copper, or of corrosion resistance steel material weldment, fitted with necessary covered handholes, air vents, oil level gauge, filling funnel with 100 mesh strainer, piping connections for filling, draining, circulating, and replenishing lines and shutoff valves or cocks as required. Tanks shall withstand 8 pounds per square inch (lb/in^2) hydrostatic pressure without failure or distortion. For class 2 and 3 windlasses, the bedplates for the power plants may be constructed to serve as the oil storage tank and replenishing tank and separate replenishing pumps provided. Class 1 windlasses shall have the piping arranged so as to permit operation of either power plant from either tank and the leveling of oil from one tank to the other.

3.4.8.7 Bends in piping shall be to a radius of not less than 5 times the nominal pipe size (nps) of the pipe.

3.4.9 Electrical requirements.

3.4.9.1 Electric motors and motor characteristics shall be in accordance with MIL-M-17060 and MIL-M-17413 as applicable (see 6.2).

3.4.9.2 Electric motor controllers and controller characteristics shall be in accordance with MIL-C-2212 and as specified (see 6.2).

3.4.9.2.1 Control of the power unit motors shall be by push button master switches, one located on the main deck and one in the windlass room below. "Start", "Stop", and "Emergency Run" elements shall be provided in the master switches.

3.4.9.2.2 A neutral interlock switch shall be provided and mounted at each hydraulic pump. It shall prevent the electric motor from being started under load and necessitate placing the pump in neutral or no-stroke position before the motor may be started.

3.4.9.3 Electric brakes with brake characteristics in accordance with MIL-B-16392 shall be provided. The brakes shall be arranged to set upon loss of power or when power is not applied, to prevent the anchor and chain from overhauling the anchor windlass. An electric brake shall be provided between the hydraulic motor and the gears. The brake shall stop the windlass upon loss of power to the electric motor or to the hydraulic motor. The electric brake shall have sufficient capacity to stop and hold 150 percent of the weight of the anchor and 60 fathoms of chain while being lowered at full speed under control of the power unit.

3.4.10 Hydraulic requirements.

3.4.10.1 Relief valves shall be provided in the pipelines adjacent to the main hydraulic pump. Relief valves shall be set to relieve at 200 percent of normal operating pressure required for hoisting an anchor from a depth of 60 fathoms or in accordance with the pump manufacturer's recommendations but not in

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excess of 200 percent.

3.4.10.2 Relief valves shall provide for relief from both sides of the hydraulic system and shall discharge to the low pressure side of the main hydraulic piping.

3.4.10.3 A replenishing pump shall be provided for delivering oil to the replenishing valves at constant pressure when forced replenishing is required. The pump shall be driven directly from the main power unit and shall deliver the oil through a 10 micrometer absolute filter. Each pump shall take its suction from the tank with a separate replenishing line with relief valves provided from the pump to replenishing valves. Excess oil from the replenishing pump shall relieve to the pump case and thence, through circulating lines, to the tank.

3.4.10.4 Suitable valves with threaded outlets shall be provided at low points in drainage piping and filling points to allow for attachment of a portable filtering unit in accordance with Drawing 53711-5381250.

3.4.10.5 All hydraulic piping, tubing fittings, and associated components shall be in accordance with MIL-STD-777. Piping shall be of sizes that will limit the maximum velocity of oil under pump pressure to 20 feet per second.

3.4.10.6 The hydraulic piping shall be so arranged and provided with valves and fittings that the oil in the hydraulic system can be filtered through an oil filter provided as part of the windlass equipment. The filter shall remove particles over 10 micrometer size and be located in the windlass room and arranged so that oil which has been drained or pumped from the system into a storage tank may be pumped back through the filter in not more than 1 hour. A separate pump shall be provided for this purpose. For class 1 windlasses, the pump shall be motor driven. An instruction plate, attached to the filter, shall contain directions for cleaning and servicing the filter.

3.4.10.7 Heat exchangers, when required, shall be shell-and-tube, fluid-to-water type and shall be of 70-30 Copper-Nickel (Cu-Ni) double tube construction in accordance with MIL-C-15730.

3.4.10.8 Hydraulic filters shall be in accordance with MIL-F-24402. Filters shall be readily accessible with sufficient clearance provided for removal of disposable type filter element. Filters shall include a by-pass relief and differential pressure indicator.

3.4.10.9 A set of special tools shall be as specified (see 6.2). Special tools are defined as those not listed in the Federal Supply Catalog. (This catalog may be consulted in the office of the Defense Contract Management Area Operations (DCMAO).) Each tool shall be indelibly marked to indicate the services for which intended.

3.5 Allowable stresses. Stress calculations for windless components shall be made (see 6.3).

3.5.1 Under rated normal working load of hoisting the anchor and its chain from 60 fathoms depth, the combined stresses in no part shall exceed 35 percent of the tensile yield point of the ductile materials used.

3.5.2 Under extreme loading such as to cause lifting of relief valves,

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the combined stresses shall not exceed 70 percent of the tensile yield point.

3.5.3 Stresses in no part of the equipment shall exceed 70 percent of tensile yield point when the capstan head is subjected to a horizontal pull equal to minimum breaking strength of the synthetic rope (see 6.2), acting at midheight of the capstan head and normal to the axis of the shaft.

3.5.4 Stresses in any part of wildcat brakes or their operating mechanisms and mountings shall not exceed 25 percent of tensile yield point under any of the operating or test conditions herein specified.

3.5.5 In stress calculations, the following strength relationship shall exist:

Design strength	Percent of tensile yield point
Direct shear	60
Torsional shear	60
Compression (bearing)	160

3.5.5.1 Other values may be used if substantiated by tests and approved by the bureau or agency concerned.

3.6 Identification plates. Identification plates and other designated marking plates shall be in accordance with MIL-P-15024 and MIL-P-15024/5. All plates including operating and lubrication charts exposed to the weather shall be fabricated from corrosion-resistant steel, and shall be type A, B, or C. Charts shall be mounted by fully bedding with adhesive and attached with corrosion-resistant steel fasteners.

3.6.1 Identification and marking. Identification and marking shall be in accordance with MIL-STD-130.

3.7 Painting and preservation procedures. Machinery and equipment shall be painted and preserved as specified herein, unless otherwise specified in referenced specifications. Interior surfaces not intended to remain bright shall be painted in accordance with (a)(1) through (a)(3). Equipment shall be painted in ambient conditions as follows: air surface temperature between 50 and 100 degrees Fahrenheit (°F); and relative humidity not greater than 80 percent (see 6.2).

(a) Painting procedures for steel and iron surfaces. Painting procedures for steel and iron surfaces shall be as follows:

(1) Unless otherwise specified (see 6.2), mask all surfaces not to be painted. These areas include: bearing surfaces that are required to move with respect to each other, such as threads, bearings, pins, and slides; mating surfaces relating to component alignment, proper seating, and mounting, such as counter bores and keyways; non-metallic surfaces, such as rubber and plastic; and all press force, drive and slip fits.

(2) Sandblast to white metal in accordance with SSPC SP5. Blast

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to achieve a paint anchor surface profile of 1 to 2-½ mils (25 to 65 micrometers).

(3) Within 2 hours of sandblasting, and while white metal conditions as defined above still exist, apply one coat, 2 to 4 mils dry film thickness (dft), of self-cure type solvent base inorganic zinc silicate primer selected from NAVSEA S9086-VD-STM-000/CH-631 applicable to surface ships, paint systems for exterior, vertical and horizontal surfaces above boottopping.

(4) Apply two coats of formula 150 (green) epoxy paint in accordance with MIL-P-24441 and MIL-P-24441/1. Mixing is to be in accordance with the paint manufacturer's instructions. The first coat shall be a mist coat to fill the porosity in the zinc silicate. The mist coat only shall be thinned 10 to 15 percent with the thinner consisting of 50-50 mixture of high flash naphta and butyl alcohol and applied to a wet film thickness (wft) of 1 to 2 mils. The second coat shall be applied to a dft of 2 to 4 mils when the first coat is between the condition "dry to recoat" and "tack-free" as defined by ASTM D 1640 (but in no case longer than 24 hours).

(5) Two coats of color no. 26270 of FED-STD-595 (haze gray), silicone alkyd enamel shall be applied to a dft of 1-½ mils minimum per coat in accordance with TT-E-490. The first coat shall be applied when the last coat of formula 150 is between the condition "dry to recoat" and "tack-free" as defined in ASTM D 1640.

(b) Painting procedures for gear oil ferrous surfaces. Painting procedures for gear oil ferrous surfaces shall be as follows:

(1) Sandblast to white metal in accordance with SSPC SP5. Blast to achieve a paint surface profile of 1 to 2-½ mils (25 to 65 micrometers).

(2) Mask all surfaces not to be painted. Unless otherwise specified (see 6.2), these areas include: bearing surfaces that are required to move with respect to each other, such as threads, bearings pins and slides; mating surfaces relating to component alignment, proper seating and mounting, such as counter bores and keyways; non-metallic surfaces, such as rubber and plastic; and all press force, drive and slip fits.

(3) Within 2 hours of sandblasting and while white metal conditions as defined in (a) still exist, apply one coat of formula 150 (green) epoxy paint to a dft of 2 mils in accordance with MIL-P-24441/1.

(4) Apply two coats of formula 152 (white) epoxy paint in accordance with MIL-P-24441/3 to a dft of 2 to 4 mils. Each coat shall be applied when the preceding coat is between the condition "dry to recoat" and "tack-free" as defined by ASTM D 1640.

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(c) Painting procedures for external surface touch-up. Painting procedures for external surface touch-up shall be as follows:

(1) Unless otherwise specified (see 6.2), areas to be touched-up include all chipped and scratched external surfaces. Prepare surfaces to be touched-up by thorough solvent cleaning and sanding with 100 grit sandpaper to roughen surface and feather edges. Mask surfaces not to be painted. Apply one coat of formula 150 (green) epoxy paint to a wft of 2 mils minimum in accordance with MIL-P-24441. Follow mixing instructions of the paint manufacturer.
(2) Apply one coat of haze gray silicone alkyd enamel in accordance with TT-E-490 to a dft of 1 to 1-½ mils minimum when the formula 150 (green) epoxy paint is between the condition "dry to recoat" and "tack-free" as defined in ASTM D 1640.
(3) Solvent clean and thoroughly dry all previously painted surfaces of the unit, part, or assembly. Mask all previously unpainted surfaces including label plates.
(4) Apply one coat of haze gray silicone alkyd enamel in accordance with TT-E-490, to a dft of 1 to 1-½ mils minimum.

3.8 Workmanship. The windlasses and components shall be free from all imperfections which might impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Shop tests.

4.2.1 Location. Shop tests shall be made in the place of manufacture.

4.2.1.1 Hydraulic units shall be tested in accordance with MIL-P-17869.

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4.2.1.2 Tests of all electrical equipment shall be made in accordance with applicable specifications.

4.2.1.3 Expansion and replenishing tanks shall be tested to 8 lb/in^2 hydrostatic pressure for tightness and rigidity.

4.3 Dock and sea tests. The following tests shall be conducted on each windlass after installation on board ship:

4.3.1 Dock tests. Each equipment shall be tested as follows:

4.3.1.1 Hydraulic system. Before conducting operating tests, the high-pressure piping system, exclusive of pumps, shall be tested with operating fluid to a pressure equivalent to 135 percent of the system design pressure. The system design pressure is the main pressure relief valves setting. The low-pressure free venting piping and supply tank of the hydraulic system shall be tested to a pressure of not less than 8 lb/in^2. Low-pressure piping and tanks which operate under a positive pressure shall be tested to a pressure equal to 150 percent of the working pressure.

4.3.1.2 Capstan static test. Each capstan head shall be tested statically by using the head as a bitt to hold a load equal to the breaking strength of the synthetic rope for 10 minutes as specified (see 3.3.2). The capstan driving mechanism will not be required to raise a weight equal to the static rating.

4.3.1.3 Test conditions. During the tests specified in 4.3.1.4 through 4.3.1.9, the equipment will not be required to operate under full load conditions for more than 1 hour out of each 4-hour period. Sufficient readings shall be taken to calculate the performance characteristics of each unit tested.

4.3.1.4 No load test. Each windlass unit shall be run for 30 minutes in each direction, with wildcats disconnected.

4.3.1.5 Normal operation test. Each windlass unit shall be tested by hoisting and lowering its anchor to 30 fathoms in 33 fathoms or more of water at an average rate of not less than 6 fathoms per minute continuously for 4 complete cycles. This test may be conducted in a lesser depth of water, provided that sufficient weights are added so that the total weight lifted is equivalent to the weight of the anchor and 30 fathoms of chain in the water. This test shall also cover inspection of the operation of the chain on the wildcat and in the chain pipe. Without exceeding the full load rating of the motor, each windlass shall be tested by hoisting a weight equal to 100 fathoms of chain plus the anchor weight (with compensatory weight added as described above) a distance of 30 fathoms at no specified speed.

4.3.1.6 Cross-over drive for class 1 electric hydraulic windlass. The units shall be tested by hoisting and lowering each anchor to 30 fathoms (or with compensatory weight added as described above) at an average rate of not less than 6 fathoms per minute, with the starboard hydraulic pump driving the port wildcat and, subsequently, the port pump driving the starboard wildcat. These tests shall be repeated a sufficient number of times to demonstrate satisfactory operation of the cross-over drive.

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4.3.1.7 Electric brake test. A weight equal to 100 fathoms of chain plus the anchor weight shall be lowered as in normal operation until the anchor is within 3 fathoms of the bottom at which time the electric motor shall be de-energized and the electric brake allowed to function. The electric brake shall stop and hold the load before the anchor grounds.

4.3.1.8 Hand brake test. Each unit shall be tested with weight added as above so that the total weight is equivalent to the weight of the anchor and 30 fathoms of chain in the water. The anchor shall be held in stowage position in the hawse pipe by the hand brake while the locking head is disconnected. The brake shall then be released and the anchor let go until 15 fathoms of chain are out or, in case the water is less than 15 fathoms deep, until the anchor is within 2 fathoms of the bottom, at which time the brake shall be applied. The brake shall stop and hold the anchor and chain within 2 fathoms travel or before the anchor grounds. Repeat the brake tests as may be necessary to demonstrate that the wildcat hand brake operates satisfactorily. When windlasses are equipped with the hydraulic power assist controls, the above brake tests shall be repeated by using these controls to demonstrate satisfactory performance.

4.3.1.9 Capstan, normal load test. Each capstan head with the wildcats disconnected and power unit operating at full speed, shall be used separately to demonstrate by tension dynamometer or by other means, that the capstan is capable of pulling its rated load at rated speed. No part of the equipment shall show any sign of permanent distortion or undue wear after these tests.

4.3.2 Deep sea tests. When the ship is on sea trials and in a depth of at least 60 fathoms of water but not exceeding 65 fathoms, anchor-handling arrangements and gear shall be tested as follows to demonstrate satisfactory operation in all respects under service conditions and sufficient data shall be recorded for this purpose:

4.3.2.1 Hoisting test. With the anchor bending shackle just outside the hawsepipe, each anchor shall be independently lowered under power until the 60-fathoms detachable link reaches the waterline, then hoisted to the stowage position in the hawsepipe. Lowering and hoisting of the anchors and chain to the 60-fathoms depth shall be at an average rate of not less than 6 fathoms per minute.

4.3.2.2 Cross-over drive test - class 1 windlasses. Each anchor shall be lowered and hoisted with the opposite power unit sufficiently to demonstrate the functioning of the cross-over connection in accordance with this specification.

4.3.2.3 Wildcat hand brake test. With the anchor in the hawsepipe and with the locking head disconnected, the anchor shall be dropped by freeing the hand brake. Anchor descent shall be stopped at 15, 30, 45, and 60 fathoms of chain out of the hawsepipe, by operating the brake handwheel without additional leverage. The brake shall stop and hold the anchor and chain in not more than 2 fathoms of travel with 15 and 30 fathoms out, in not more than 3 fathoms with 45 fathoms out and in not more than 4 fathoms with 60 fathoms out. When windlasses are equipped with the hydraulic power assist controls, the above brake tests shall be repeated by using these controls to demonstrate satisfactory performance.

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4.3.2.4 Electric brake test. Electric brakes shall be tested while lowering anchor at the maximum speed under power (see 6.3). When 60 fathoms of chain have been let out, the electric motor shall be de-energized and the electric brake allowed to function. The brake shall stop and hold the anchor and chain before anchor grounds.

4.3.2.5 Readings to be taken. The following readings shall be taken during the hoisting and cross-over tests:

(a) Revolutions per minute (rpm) of wildcat shaft.
(b) Volts, amperes, kilowatt input, and power factor if applicable.
(c) Oil pressures and temperatures.
(d) RPM of electric motor.
(e) Time.
(f) Hoisting and backing out speeds.
(g) Performance of hand brakes.
(h) Temperatures of gearing and bearings before, during, and after runs.
(i) Performance of electric brakes.

4.4 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.6.)

5.1 Preservation, packing, and marking. Equipment, detached components, and data shall be preserved level A, C, or commercial, packed level A, B, C, or commercial, as specified, and marked in accordance with MIL-M-3184 (see 6.2). In addition, for Navy acquisitions, the following applies:

(a) Navy shipboard stowage fire-retardant requirements.

(1) Lumber and plywood. Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping container and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Level A and B	-	Type II - weather resistant. Category 1 - general use.
Level C	-	Type I - non-weather resistant. Category 1 - general use.

(2) Fiberboard. Fiberboard used in the construction of class-domestic, non-weather resistant fiberboard, and cleated fiberboard boxes shall meet the flame spread index and the specific optic density requirements of PPP-F-320 and amendments thereto.

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5.2 Technical manuals. Technical manuals which accompany shipments shall be packaged in a transparent waterproof plastic bag, minimum 4 mil thick. Closure shall be by heat sealing. Technical manuals shall not be placed with any flexible sealed barrier enclosing components. The copies of the manual shall be placed in the shipping container housing the main unit. Packing lists shall indicate which container contains the technical manuals and shall state the approximate location therein. The manuals shall be readily accessible when the container is opened. Technical manuals, when shipped in bulk quantities, shall not be individually wrapped, but shall be packed in accordance with the requirements of the applicable technical manual specifications.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. It is intended that this specification be used for reference in ship specifications and for direct reference in contracts or orders for windlasses. The intent is to set forth the requirements within which the equipment must operate satisfactorily and reliably.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Class required (see 1.2).
(c) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) Maximum permissible size and weight of windlass, if required (see 3.2.4.3).
(e) Speed limiting device, if required (see 3.2.4.7).
(f) Size of chain (see 3.2.5).
(g) Size and capacity of capstan head, size, and breaking strength of the rope (see 3.2.5.3 and 3.5.3) for which the head shall be designed.
(h) Size of anchor chain and weight of anchor (see 3.3.1).
(i) Whether the bedplate and chain fairleads are required (see 3.4.3.1).
(j) Whether chain pipe deck bolsters shall be fitted with portable watertight bucklers (see 3.4.3.1).
(k) Characteristics of motor to be supplied (see 3.4.9.1).
(l) Characteristics of motor controller (see 3.4.9.2).
(m) Special tools, if required (see 3.4.10.9).
(n) Painting and preservation procedures required (see 3.7).
(o) Level of packaging and packing (see 5.1).
(p) Navy fire-retardant material (see 5.1(a)(1)).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided

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and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.2	DI-DRPR-81000	Product drawings and associated lists	----
3.5	DI-GDRQ-80650	Design data and calculations	----
4.3.2.4	DI-NDTI-80809	Test/inspection report	----

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.5.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.6 Bid data. Bid data should include the following:

(a) An approximate dimensioned general arrangement drawing showing the windlass and component parts.
(b) A complete description of the windlass and drive including gear reductions.
(c) Make, size, and type of hydraulic units.
(d) Estimated weight of complete windlass unit.
(e) Horsepower rating and frame size of proposed electric motor.
(f) Manufacturer's identification or proposed electric controller and master switch.

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6.7 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.8 Subject term (key word) listing.

Brakes, electric
Capstan
Chain
Wildcat

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 3950-N048)

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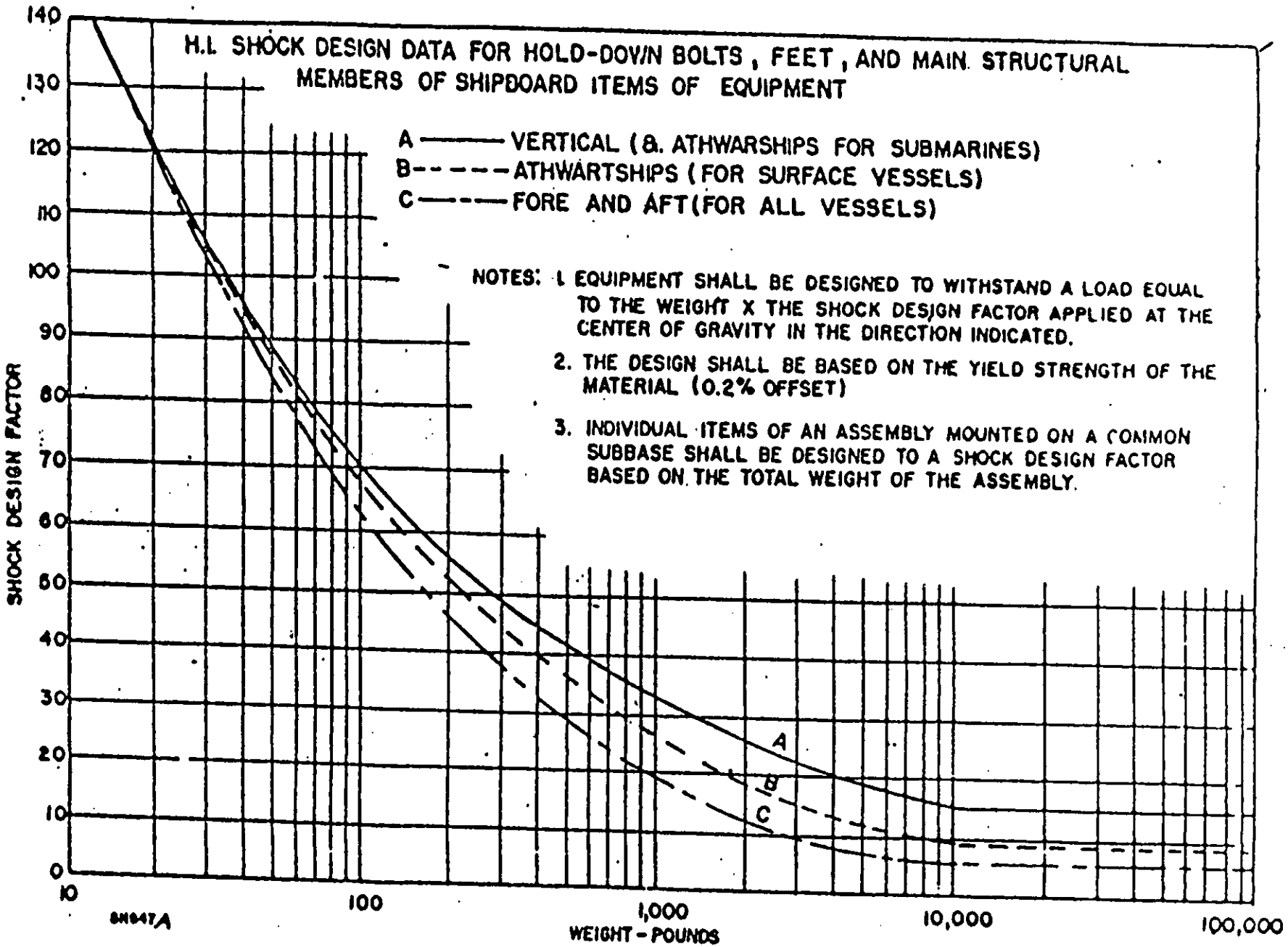


FIGURE 1. Shock design data.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-W-19623B(SH)	1994 APRIL 11

3. DOCUMENT TITLE
WINDLASS: ANCHOR, ELECTRIC-HYDRAULIC, VERTICAL AND NAVY TYPE, NAVAL SHIPBOARD

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
TECHNICAL ENGINEER: MR. JOHN COON (NAVSEA 03W23)	(1) Commercial (703) 602-21844	(2) AUTOVON 332-1844
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command NAVSEA 03R42, 2531 Jefferson Davis Hwy Arlington, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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